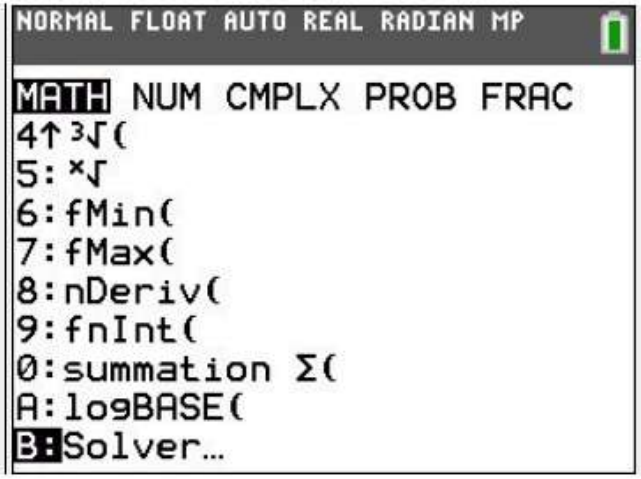
Lesson 30: Arc Length of Plane Curve and Parametric Curve

TI-84 Calculator fnInt Feature



Numerical Integration on TI-84:

Press [MATH]

Select fnInt

Input the function, variable, and lower/upper bounds

Example: To find $\int_0^2 x^2 \cdot dx$

Input fnInt(X^2, X, 0, 2). for older TI-84

Input $\int_0^2 (x^2)dx$ for newer TI-84

1) Let $f(x) = (x^2 + 1)^{3/2}$

Find the arc length of the graph of $f(x)$ on the interval $[0,2]$.

a) $f'(x) =$ ___?___

b) $s = \text{arc length} = \int_a^b \sqrt{1 + [f'(x)]^2} \cdot dx =$ ___?___

Hint: Use TI-84 fnInt feature to evaluate the definite integral.

Answers

a)

b)

2) Let $f(x) = \ln(3x^3 + 1)$

Find the arc length of the graph of $f(x)$ on the interval $[0,2]$.

a) $f'(x) =$ ___?___

b) $s = \text{arc length} = \int_a^b \sqrt{1 + [f'(x)]^2} \cdot dx =$ ___?___

Hint: Use TI-84 fnInt feature to evaluate the definite integral.

Answers

a)

b)

3) Let $f(x) = \dfrac{1}{x+1}$

Find the arc length of the graph of $f(x)$ on the interval $[0,2]$.

a) $f'(x) =$ ___?___

b) $s = \text{arc length} = \int_a^b \sqrt{1+[f'(x)]^2} \cdot dx =$ ___?___

Hint: Use TI-84 fnInt feature to evaluate the definite integral.

Answers

a)

b)

4) Let $g(y) = 3y^2 + 2$

Find the arc length of the graph of $g(y)$ on the interval $y \in [0,2]$.

a) $g'(y) =$ ___?___

b) $s = \text{arc length} = \int_{c}^{d} \sqrt{1 + [g'(x)]^2} \cdot dy =$ ___?___

Hint: Use TI-84 fnInt feature to evaluate the definite integral.

Answers

a)

b)

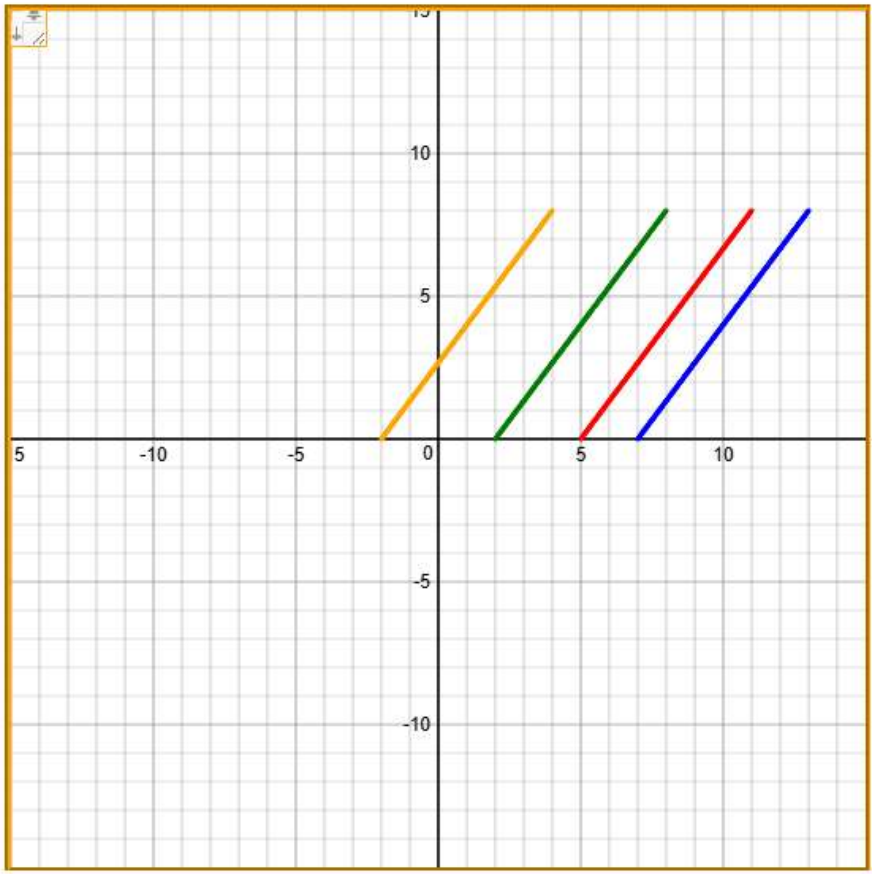
5) Parametric Equations: $x = 3t + 2, \quad y = 4t \qquad 0 \le t \le 2$

a) Graph parametric equations $x = 3t + 2, \quad y = 4t \qquad 0 \le t \le 2$

(multiple choice). Choose the correct graph (blue, red, green, or orange)

b) $s = \text{Arc Length} = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt =$?

Hint: Use TI-84 fnInt feature to evaluate the definite integral.



Answers

a)

b)

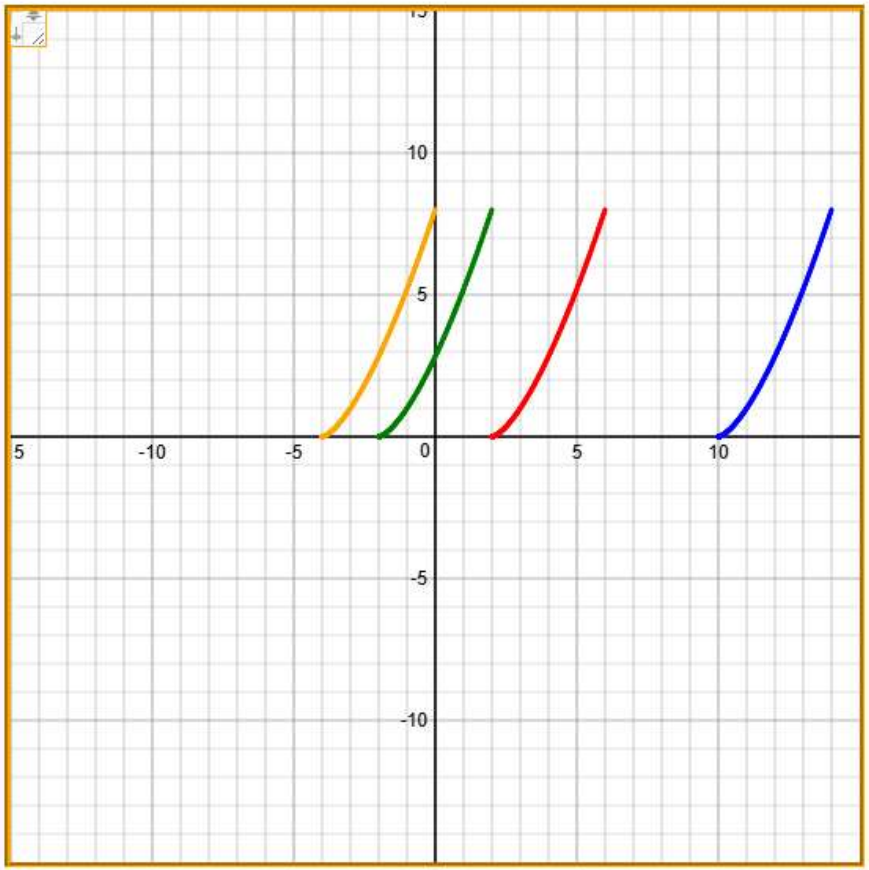
6) Parametric Equations: $x = t^2 + 2, \quad y = t^3 \qquad 0 \le t \le 2$

a) Graph parametric equations $x = t^2 + 2, \quad y = t^3 \qquad 0 \le t \le 2$

(multiple choice). Choose the correct graph (blue, red, green, or orange)

b) s = Arc Length = $\int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt =$?

Hint: Use TI-84 fnInt feature to evaluate the definite integral.



Answers

a)

b)

7) Parametric Equations: $x = 3\cos t + 2,\ \ y = 4\sin t \qquad 0 \le t \le \pi/3$

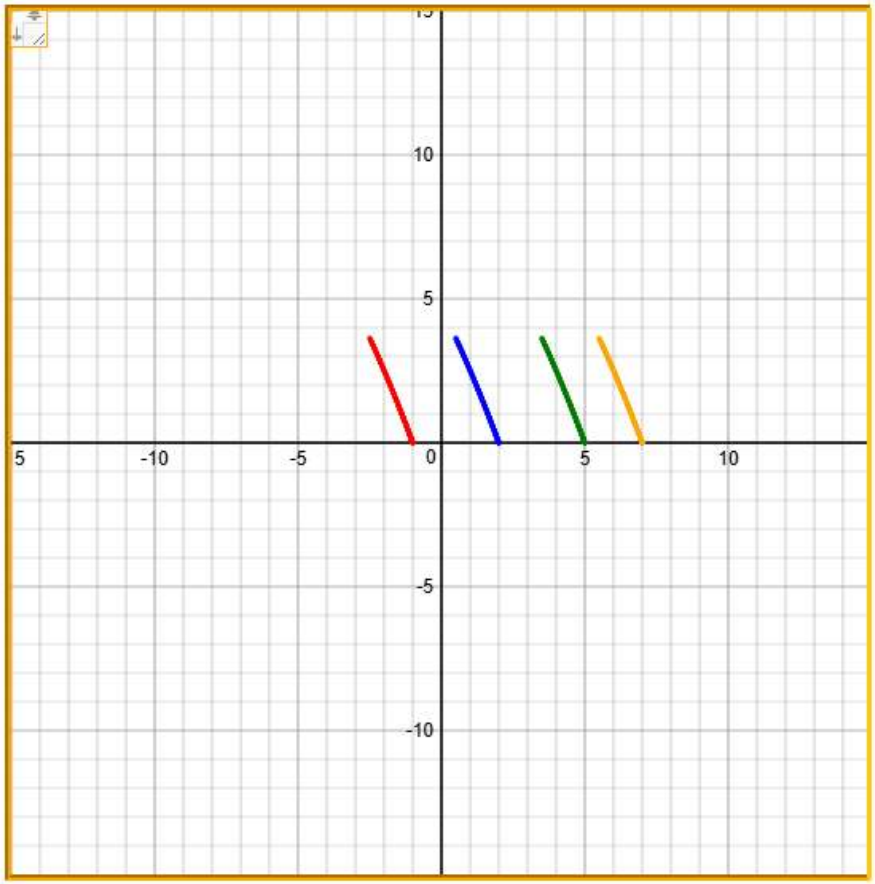
a) Graph parametric equations $x = 3\cos t + 2,\ \ y = 4\sin t$

$$0 \le t \le \pi/3$$

(multiple choice). Choose the correct graph (blue, red, green, or orange)

b) $s = \text{Arc Length} = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt =$?

Hint: Use TI-84 fnInt feature to evaluate the definite integral.



Answers

a)

b)

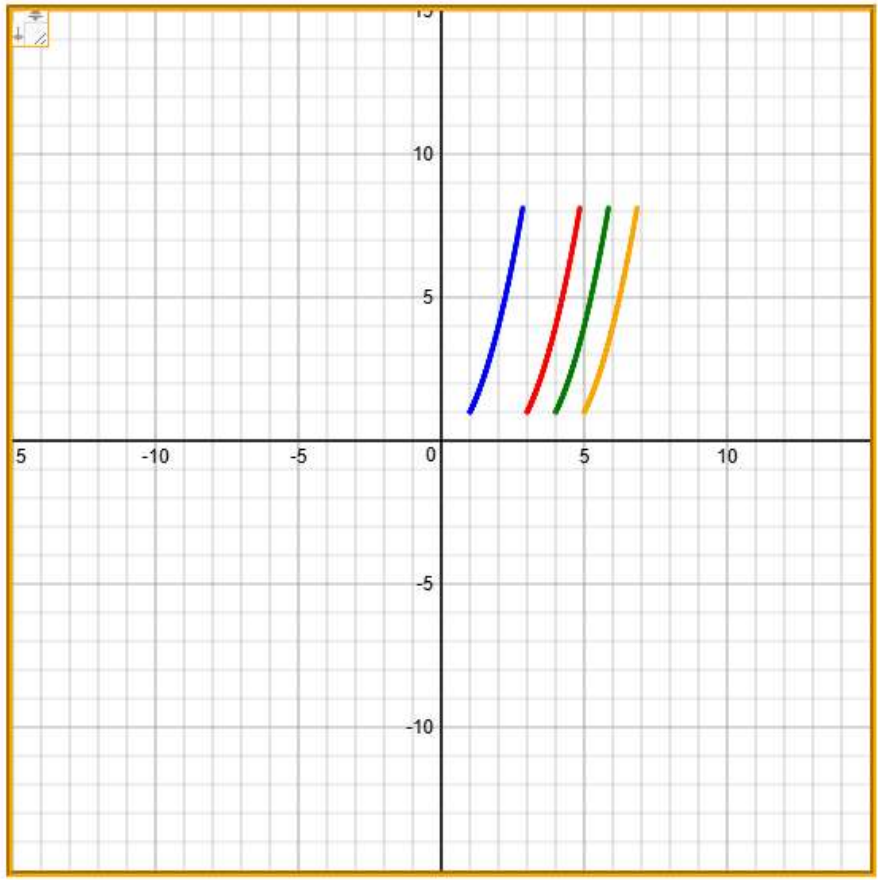
8) Parametric Equations: $x = e^t, \ y = e^{2t} \qquad 0 \le t \le 1$

a) Graph parametric equations $x = e^t, \ y = e^{2t} \qquad 0 \le t \le 1$

(multiple choice). Choose the correct graph (blue, red, green, or orange)

b) $s = \text{Arc Length} = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt =$?

Hint: Use TI-84 fnInt feature to evaluate the definite integral.



Answers

a)

b)